Decomposing the GP Advantage in Offline MBO

Anonymous submission

Abstract

Offline model-based optimization ships a surrogate and an optimizer together and reports one number, leaving attribution between them open—as the subfield's own 2026 survey states. We cross both axes under one shared protocol, decompose the variance, and name five confounds specific to this comparison. The audit genre's shape is not ours. Correcting the two protocol confounds, target scaling and the candidate/oracle rule, does not shrink the reported surrogate advantage but strengthens it, moving η^2_{surr} from a published 0.37 to 0.405 $[0.290, 0.556]$—a direction we find no precedent for in the ML audit literature, where corrected effect sizes shrink; the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds, not a property of de-confounding in general. The four corner intervals are not resolvable at $n{=}7$ tasks; firm where they are not is the β-sensitivity, η^2_{surr} rising from 0.184 $[0.085, 0.354]$ at $\beta{=}0$ to 0.406 $[0.285, 0.564]$ at $\beta{=}2$, and doubling again with search budget. The mechanism is narrowed by elimination. With pessimism removed the GP family retains 61% of its advantage (0.319 $[0.196, 0.460]$ against 0.525 $[0.406, 0.614]$), so pessimism amplifies a mean-quality base rather than creating it; sweeping member width $10.7\times$ at fixed K leaves the gap unchanged; and the ensemble, though the more accurate surrogate in-distribution, still loses. A pre-registered bidirectional manipulation then refutes the account we had published: smoothing the ensemble's mean by up to 98% widens the gap rather than closing it. Seven explanations are eliminated and none confirmed. The seventh measures what the others assumed: the ensemble's optima are worse in true oracle value (-1.406 $[-2.803, -0.375]$) yet sit no further off-support and are no more inflated. What differs is which off-distribution maximizers a surrogate's mean admits, not how rough it is or how far out it reaches. Under a matched query budget the optimizer main effect is $\eta^2_{\text{opt}}{=}0.038$ $[0.003, 0.123]$, eight times the published value: established at high budget, underpowered at low. On Design-Bench we detect no difference at this power, which at $n{=}7$ tasks is not a demonstration of equivalence, partly because some surrogate cells are frozen and cannot move.

1 Introduction

Offline model-based optimization (MBO) seeks a design x maximizing an expensive black-box f from a fixed dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, with no further queries to f during search (Trabucco et al. 2022; Kim et al. 2025). The dominant recipe fits a surrogate $\hat{f}$ and optimizes it; because optimization pushes designs into regions where $\hat{f}$ is unconstrained, the surrogate is typically made conservative—by a learned regularizer (Trabucco et al. 2021) or by a pessimistic acquisition such as the lower confidence bound (LCB) $\mu(x) - \beta\sigma(x)$ (Srinivas et al. 2012). A common intuition is that Gaussian-process LCB (GP-LCB) beats neural-ensemble LCB at low dimension because the GP posterior is better calibrated. But the two pipelines differ in *two* coupled factors, not one: the surrogate class (GP posterior vs. ensemble disagreement) and the acquisition optimizer, since GPs are usually optimized by perturbation or quasi-Newton restarts and neural surrogates by gradient ascent on x. Every method in this lineage ships both at once and reports one number.

The attribution question is open, by the field's own account. The subfield's 2026 survey states that existing benchmarks "often emphasize overall optimization performance without clarifying whether observed gains stem from superior surrogate modeling, improved optimization strategies, or mere chance" (Kim et al. 2025). That is not our characterization of a gap; it is the field certifying that the attribution is unresolved. Resolving it takes a design nobody has run: both axes manipulated against each other under one shared protocol, with the outcome variance decomposed between them. The systematic studies that exist fix one axis and vary the other—surrogate-class comparisons hold the optimizer constant (Tan et al. 2025; Li, Rudner, and Wilson 2024), the conservative-surrogate lineage holds gradient ascent fixed while varying the regularizer (Trabucco et al. 2021)—and a design that fixes a factor cannot estimate that factor's effect. This paper runs the crossed surrogate$\times$optimizer factorial and reports the two-way variance decomposition. We know of no prior instance in offline MBO.

The contribution. The instrument, and the confound set that running it exposed. The one-way analogue over model class is owned by fANOVA, computed at a single fixed search run (Hutter, Hoos, and Leyton-Brown 2014); the closest crossed grid reports descriptive metrics in online BO (Liang et al. 2021); and the closest two-axis design names fANOVA and *declines* it, on the grounds that one-factor-at-a-time analysis cannot detect interactions (Moosbauer et al. 2022). But an instrument inherits every confound in the grid it measures, and building the crossed grid surfaced five specific to offline-MBO surrogate comparison, each traceable to a line of the scoring path. The contribution is the composition: the five confounds, the protocol that removes them, and what the decomposition says once they are gone.

The genre-shape is prior work. The second half of that—name unreported implementation choices, give the protocol that removes them, show the ranking move—is an established genre, and we claim no credit for its shape. Its canonical instances are well known: reproducibility and statistical audits of deep RL (Henderson et al. 2018; Agarwal et al. 2021), the neural-recommender audit (Ferrari Dacrema, Cremonesi, and Jannach 2019), the metric-learning reality check (Musgrave, Belongie, and Lim 2020), and the large-scale GAN comparison (Lucic et al. 2018). One of them appeared at this venue. We say so early and by name: a compositional residual is only credible from authors who state plainly which parts are not theirs.

Contributions.

1. **Five named confounds and a removal protocol** (Section 3), each with a code-level trace and a signed effect on η^2_{surr}.
2. **An audit that strengthens its own headline** (Section 4). Correcting the two *protocol* confounds moves η^2_{surr} from a published 0.37 to 0.405 $[0.290, 0.556]$—a direction with no precedent we could find in the ML audit literature, where audits normally shrink (Melis, Dyer, and Blunsom 2018); the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds, not a property of de-confounding in general. The corner intervals are not resolvable at $n{=}7$ tasks, so we lead with the β-dependence, which is.
3. **A mechanism narrowed by elimination** (Section 5), applied and diagnosed rather than discovered (Liu et al. 2020; Fan et al. 2024): seven controls remove σ, width, budget, held-out accuracy, mean roughness, premise coverage, and distance-to-support with surrogate inflation—the last three by pre-registered manipulations that refute accounts of our own.
4. **A Design-Bench null with its power stated** (Section 6): no detectable difference at $n{=}7$ tasks, below the threshold for the omnibus we run (Demšar 2006), reported as such and never as equivalence (Agarwal et al. 2021)—partly because some surrogate cells are frozen at a fixed dataset reference and cannot move at all.

Scope. No field reversal: that the statistical model often matters more than the acquisition heuristic is the organizing doctrine of the standard Bayesian-optimization review (Shahriari et al. 2016) and is a decade old, and our optimizer-axis result falsifies the *local* premise of one recent paper (Chemingui et al. 2024) and nothing wider. We claim no mechanism discovery, no equivalence on Design-Bench, and no result about ensemble size; Section 7 states each in full.

2 Background and Related Work

LCB, pessimism, and coverage. GP-LCB $\mu - \beta\sigma$ has regret guarantees under GP assumptions (Srinivas et al. 2012); deep ensembles (Lakshminarayanan, Pritzel, and Blundell 2017) provide a cheap uncertainty proxy but no calibration guarantee. The pessimism principle behind conservative offline methods is that a valid lower bound prevents the optimizer exploiting model error (Jin, Yang, and Wang 2021); we make that premise measurable. Conformal methods give distribution-free coverage, including inside the Bayesian-optimization loop (Angelopoulos and Bates 2023; Stanton, Maddox, and Wilson 2023); we use coverage of the LCB premise only as a *measurement*, taken where the optimizer actually operates, and Section 5 shows it is separable from optimization outcome—a caution against treating it as the mechanism.

Offline MBO and its evaluation. The conservative-surrogate lineage runs from Conservative Objective Models (Trabucco et al. 2021), which add a CQL-style regularizer (Kumar et al. 2020), through generative and ranking-based surrogates (Brookes, Park, and Listgarten 2019; Krishnamoorthy, Mashkaria, and Grover 2023; Tan et al. 2025); all hold the search rule fixed while varying the model. Optimizer-as-contribution work (Chemingui et al. 2024) argues the converse but proposes a method rather than decomposing, and Design-Bench (Trabucco et al. 2022) standardizes tasks and, by fixing a protocol, documents the confounded status quo we dissect. Each canonical audit in the reality-check genre pairs a critique with a reusable tool (Henderson et al. 2018; Agarwal et al. 2021; Balduzzi et al. 2018), and in offline optimization SOO-Bench (Zhu, Qian et al. 2025) stress-tests optimizer *stability* rather than attribution. Our study targets the surrogate×optimizer confound, which none of the two-axis precedents the introduction names (Hutter, Hoos, and Leyton-Brown 2014; Liang et al. 2021; Moosbauer et al. 2022) decomposes.

3 Experimental Grid and Confounds

The grid. Three surrogates $\times$ three acquisition optimizers $\times$ 7 synthetic tasks (Branin-2D through Griewank-30D, dataset drawn once at seed 0) $\times$ 30 seeds, plus 7 Design-Bench tasks at 16 seeds. **Surrogates:** a $K{=}5$ ensemble of two-hidden-layer ReLU MLPs (width 96, MSE, 35 epochs, Adam lr 3×10^{-3}), with μ, σ the member mean and standard deviation; an *exact GP* (BoTorch `SingleTaskGP` at its default covariance module, fit by marginal likelihood on a score-biased subsample, $N_{\max}{=}800$); and a *sparse variational GP* (SVGP; 128 inducing points, ARD Matérn-$5/2$, 250 ELBO steps, $N_{\max}{=}2000$). **Optimizers:** 100 Adam steps on x (lr 0.05, box-clipped); 5 rounds of Gaussian hill-climbing; and CMA-ES. All three surrogates are differentiable, so all nine cells are realizable. Every cell maximizes the LCB $\mu(x) - \beta\sigma(x)$ at $\beta{=}2$ and is scored by the ground-truth oracle on its 128 returned designs at the 100th percentile. Per task we min–max normalize the nine cell means and take a two-way ANOVA; η^2_{surr} and η^2_{opt} are the fractions of normalized-score variance carried by the two main effects, with task-and-seed hierarchical bootstrap intervals (B=10,000).

Two details of that description are themselves engine state: the exact GP's default covariance is ARD RBF under the BoTorch pinned for the synthetic grid but ARD Matérn-$5/2$ under the older version stamped on the Design-Bench runs, and the GPs alone spend per-run marginal-likelihood budget, which a matched-tuning arm freezes to zero for every surrogate (the surrogate effect retains 76% under that control). Neither is recoverable from a method description that does not stamp its engine.

Against that grid we name five confounds. Each is a specific line of the scoring path, not a judgment call, and each carries a signed effect (Table 1).

Confound 1: target scaling. The ensemble regresses on raw targets spanning -2613 to $+36$ while both GP surrogates z-score theirs (`mbo.py:36-37,130-138` against `mbo.py:255,311-312`). Correcting this alone moves η^2_{surr} from 0.367 $[0.254, 0.559]$ to 0.283 $[0.186, 0.444]$. This is a measurement confound, not a bug report: the prior analysis states a min–max normalization that exists only in the analysis path, never in training, and the gap between stated and actual protocol is the finding.

Confound 2: candidate/oracle protocol. Two of three optimizers propose 256 designs and report the oracle-selected best 128, while the third proposes 128 (`mbo.py:384-389,188,198-201,292,392-394`). The reported percentile is therefore two different estimands in one column. Correcting this alone moves η^2_{surr} from 0.367 to 0.450 $[0.312, 0.649]$. This is the strongest of the five: the protocol the prior analysis declares it holds identical is the identifiability license for the whole comparison, and the license is false.

Confound 3: ensemble size K. K enters as a confound to be controlled, never as a mechanism, and has two halves that must be read together. It is *sensitive*: η^2_{surr} is 0.326, 0.366, 0.408, 0.389 at $K = 2, 3, 5, 10$, peaking at exactly the K=5 the field fixes by convention (Lakshminarayanan, Pritzel, and Blundell 2017). And it does *not* reverse: at K=2 the ensemble's normalized cell marginal is 0.283 against the exact GP's 0.767 and the SVGP's 0.739, so our own pre-registered kill criterion—"if the ensemble's marginal at K=2 exceeds the GP's, the surrogate main effect is a K artifact and the headline reverses"—did not fire. We also report that the figure this criterion was written against, a task-normalized ensemble score of 0.95 at K=2 in the prior supplement, *does not reproduce* on the audited engine, which gives 0.283 at that cell. (These two 0.283s are unrelated quantities: Confound 1's is a variance-explained statistic, this one is a normalized cell marginal. Neither corroborates the other.) Our sweep runs over $K \in \{2, 3, 5, 10\}$ and is therefore disjoint below the $K \in \{5, 10\}$ range over which ensemble surrogates were found robust (Li, Rudner, and Wilson 2024), and the decline with K runs against the direction reported for ensemble quality (Abe et al. 2022)—both further reasons to treat K as a nuisance coordinate, not a result.

Table 1: The five confounds. Signed effect is on η^2_{surr} (synthetic, n=7, 30 seeds) except Confound 5, which acts on η^2_{opt}. The two *protocol* confounds move the headline against the ensemble-size confound; the net is Section 4.

Confound	What it breaks	Signed effect
1. Target scaling	ensemble trains on raw y	$0.367 \to 0.283$
2. Candidate/oracle	two estimands in one column	$0.367 \to 0.450$
3. Ensemble size K	headline fixed at the peak	0.326–0.408
4. β–σ match	per-task, not aggregate	ratio 0.07–1.44
5. Query budget	optimizer = search effort	$0.005 \to 0.038$

Confound 4: β–σ mismatch. We registered that a shared β=2 delivers different effective pessimism to different surrogate classes, with the kill criterion "if the ratio is $\sim$1, the acquisition was matched and this confound does not exist." *That kill fired.* The median $\sigma_{\text{GP}}/\sigma_{\text{ens}}$ is 1.19 at matched K=5 (1.21 pooled over K), so the fifth unmatched hyperparameter does not exist grid-wide, and we withdraw the aggregate claim. What survives is per-task: the ratio spans 0.07 on Branin-2D to 1.44 on Styblinski-5D and Rastrigin-15D, a 20-fold spread that a single shared multiplier cannot track. The general principle that a fixed multiplier on an uncalibrated interval is not comparable across model classes is owned by the interval-validity literature (Dewolf, De Baets, and Waegeman 2022) and its offline-RL instance (Ghasemipour, Gu, and Nachum 2022); we claim only the offline-MBO acquisition-comparison application, in its narrowed per-task form.

Confound 5: search-intensity budget. Surrogate-query budgets are unmatched by $11.8\times$ across the optimizer axis. The counts are *measured* by a counting proxy rather than derived: gradient 51,456, perturbation 4,352, CMA-ES median 6,528 (range 932–6,536). Three corrections belong here. The previously estimated "$6\times$–$59\times$" spread was never measured, and CMA's nominal budget of 3000 is a cap that rarely binds because the library stops on its own tolerances first. The proposal-count half of this confound is already removed by Confound 2, so the surviving inequality is in surrogate queries alone and does not double-count. And the trust region flagged in our own audit is inactive in the main grid, so equalizing queries isolates budget. Matching every optimizer at Q=51,456 (achieved 51,200 / 51,456 / 51,472; 0 of 630 cells more than 5% off target) moves the optimizer main effect from a published η^2_{opt} of 0.005 to 0.038 $[0.003, 0.123]$—an eightfold understatement, which we disclose. What that magnitude licenses is deferred to Section 7.

4 De-Confounding Protocol and Results

The protocol. Standardize the ensemble's regression targets. Equalize the candidate proposal count across optimizers and impose one selection rule, so the oracle never chooses the reported set. Hold K, β and the query budget fixed and *report them.* Stamp the engine—library versions, flags, commit—with every number. Switching the first two on and off gives four corners (Table 2).

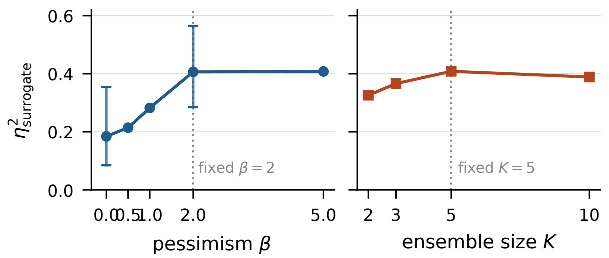

Figure 1: $\eta^2_{\text{surrogate}}$ against β (left, bootstrap 95% intervals at the measured endpoints) and against K (right), with the conventionally fixed operating point marked. The headline is reported at the maximum of both curves, and the ranking reverses nowhere on either.

The correction strengthens the surrogate effect. Audits in this genre usually shrink the effect they audit (Melis, Dyer, and Blunsom 2018). This one strengthens it. Correcting the two measurement and protocol confounds moves the rewarded surrogate advantage from a published η^2_{surr} of 0.37 *up* to a corrected 0.405 $[0.290, 0.556]$; the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds and not a property of de-confounding in general. We searched the ML reality-check and reproducibility literature for a de-confounding audit reporting a corrected variance-explained statistic above its own published value and found none, which is the claim in its exact scalar form. Two partials must be pre-empted. An ImageNet replication reports a relative slope above one—1.69 on CIFAR-10, 1.11 on ImageNet—but that is a relative slope, not a variance-explained scalar audited against its own prior value (Recht et al. 2019); and stratified reanalysis of deep-RL results moves effects both ways (Agarwal et al. 2021). Nor is the direction statistically surprising: a removed confound can as easily have been suppressing an effect as manufacturing one (Bressan 2019). The claim is that it is unprecedented as an instance, not anomalous as a statistic.

Corner resolution at n=7 tasks. We pre-registered that it would not be, and the prediction confirmed. The four corner intervals have widths 0.26–0.34 against a corner range of only 0.167, and the corrected corner's interval $[0.290, 0.556]$ contains every other corner's point estimate. So 0.405 never appears in this paper without its interval, and no corner is asserted to differ from another. The bootstrap is validated against the published precision: our uncorrected corner's interval width, 0.305, reproduces the published 0.32.

The β-dependence is the firm result. η^2_{surr} rises monotonically with pessimism—0.184 $[0.085, 0.354]$, 0.214, 0.282, 0.406 $[0.285, 0.564]$, 0.408 at $\beta = 0, 0.5, 1, 2, 5$—and the β=0 and β=2 intervals barely overlap, where the four corner intervals do not separate at all (Figure 1). This is the axis the seven tasks come closest to separating, and we lead the decomposition with it for that reason. It is budget-dependent as well, more than doubling from 0.243 at Q=4,352 to 0.526 at Q=51,456. The *magnitude* of any headline η^2_{surr} is therefore a joint artifact of four chosen operating-point coordinates—K, β, the query budget, and the engine corner—so a headline that does not state all four is not reproducible. This strengthens the decomposition rather than weakening it, because the *direction* is invariant to all four: across every β, K, budget and corner we ran, the GP marginals sit at 0.75–0.85 and the ensemble's at 0.24–0.36, never crossing. Only the size of the gap moves.

Table 2: Four-corner decomposition (synthetic, 7 tasks, 30 seeds, B=10,000 bootstrap). X1 standardizes the ensemble's targets; X3 equalizes the candidate/oracle protocol. The intervals overlap heavily: corner differences are *not* resolvable at n=7.

X1 / X3	η^2_{surr} [95% CI]	η^2_{opt}	η^2_{inter}	Friedman p
off / off	0.367 $[0.254, 0.559]$	0.013	0.165	6.1×10^{-5}
on / off	0.283 $[0.186, 0.444]$	0.036	0.146	1.7×10^{-3}
off / on	0.450 $[0.312, 0.649]$	0.006	0.152	4.1×10^{-5}
on / on	$\mathbf{0.405}$ $[0.290, 0.556]$	0.005	0.160	8.1×10^{-4}

5 Isolating the Source of the Gap

Distance-aware uncertainty in GP-like models is established (Liu et al. 2020), as is the reading of a UCB-style acquisition as local search (Fan et al. 2024). What follows applies and diagnoses those results in the offline setting. We ran seven controls, each capable of explaining the GP–ensemble gap, and none does. Two were built to confirm accounts of our own and refuted them instead.

σ is a distance signal, not an error signal. The ensemble's σ correlates with pointwise absolute error at only $\rho \approx 0.07$, but with k-NN distance to the training data at $\rho \approx 0.26$—three to four times larger, positive on all seven tasks, and above 0.28 on the four mid- and high-dimensional ones. The prior analysis concluded σ was uninformative by measuring it against the wrong target; this is the corrected measurement, bounded by prior work on distance-aware uncertainty (Liu et al. 2020; van Amersfoort et al. 2020).

Elimination 1: it is not σ. With pessimism removed entirely (β=0, pure posterior-mean maximization), 61% of the GP's advantage survives and the interval excludes zero: the marginal gap is 0.319 $[0.196, 0.460]$ at β=0 against 0.525 $[0.406, 0.614]$ at β=2. Both conditions are read on one β-invariant normalizer, a per-task min–max fit once over the pooled cell means of both and refit inside each bootstrap resample. The pessimism increment is itself significant—0.203 $[0.007, 0.396]$, p=0.020—so the statement is base-plus-amplification: pessimism *amplifies* a mean-quality base rather than creating it, and any claim that the advantage is independent of pessimism is refuted by our own data. A per-task z-score cross-check gives 0.904 against 1.473 sd—the same 0.61 ratio—so this is not a min–max artifact. Our pre-registered collapse criterion passes less comfortably than earlier per-β figures implied: 0.319 against a threshold of 0.263, a margin inside the bootstrap interval.

Elimination 2: it is not finite width. The gap could be an under-parameterization artifact: wide networks approach GP behaviour (Jacot, Gabriel, and Hongler 2018; Lee et al. 2018, 2019) and finite networks show spectral bias (Rahaman et al. 2019). Our K-sweep does *not* test that—K is cardinality, not per-member capacity, and offering it here would be a category error—so we ran the ablation that does. At fixed K=5, sweeping per-member width over a $10.7\times$ range leaves the gap unchanged: 0.480 $[0.365, 0.576]$ at w=96 against 0.476 $[0.208, 0.647]$ at w=1024, 99.1% retained, shrinkage -0.006 $[-0.210, 0.161]$. The curve is flat with noise and non-monotone—it dips to 0.336 at w=256 and returns—not a decay. Two limits are ours. The interval widens monotonically with w ($0.211 \to 0.439$), making w=1024 the least precise point on the curve, so the supported claim is that the gap *does not close*, not that it is identical there. And the sweep stops at 1024 while the kernel limits are asymptotic, so this answers the objection at practical widths and makes no asymptotic claim. As a validity check, w=96 reproduces the incumbent grid bit-exactly for the gradient and CMA cells on all seven tasks.

Elimination 3: it is not predictive accuracy. The ensemble is the *more* accurate surrogate. Its held-out normalized RMSE on a 20% split never seen by the grid is below the GP's at every width—0.445, 0.405, 0.393, 0.388 at $w = 96, 256, 512, 1024$ against the GP's 0.479—winning in 26 of 28 (task, width) cells and on all seven tasks at $w \geq 512$. And it still loses the optimization comparison by roughly 0.48. The registered form was narrower, and we report it as primary: read only at the two cells where ensemble and GP RMSE are statistically indistinguishable (Styblinski-5D at w=256, difference CI $[-0.021, 0.028]$, and at w=512, $[-0.043, 0.010]$), the mean gap is still 0.375. The 7/7 result is a post-hoc strengthening, labelled as one. Held-out NLL is not usable as a second accuracy axis—the GP's is two orders of magnitude worse—because that is a calibration finding, orthogonal to this one. *The more accurate surrogate is the one that loses*, which removes predictive error from the candidates.

Elimination 4: it is not budget, and more budget makes it worse. Equalizing surrogate queries does not close the gap. The composition of that rise matters: the ensemble's marginal *falls* as budget rises ($0.361 \to 0.240$) while both GPs rise ($0.755 \to 0.849$ and $0.713 \to 0.794$). More search pressure finds more of the off-distribution maxima the ensemble's mean admits. This was not what the budget arm was designed to test, so it is an observation, not a pre-registered result, and corroborates a mechanism rather than any η^2 magnitude.

Eliminations 5 and 6: it is not how rough the mean is, and it is not premise coverage. Our previous account named the roughness of the surrogate's mean as the axis. We tested it bidirectionally under pre-registration and it failed. Penalizing the ensemble's mean gradient cuts its measured roughness by up to 98.0% (spectral normalization, 90.3%). The gap did not close; it *widened* for every variant (base 0.306 against 0.322–0.552), no closing interval excludes zero, and the Lipschitz-bounded ensemble is markedly worse in score ($0.657 \to 0.537$ at w=96, $\to 0.346$ at w=1024). A self-test planting synthetic ground truths confirms the analyzer returns the opposite verdict when smoothing genuinely closes a gap. Smoothing fixes the *premise* and not the outcome—own-proposal coverage rises monotonically to 0.998 under the strongest penalty with no effect on the gap—so the coverage proxy is separable from what it was meant to explain, corroborating on a manipulated axis the weak $\rho(\text{coverage}, \text{score}) = 0.19$ we measure across cells. The complementary arm, roughening the GP, is void: no kernel setting raised the fitted mean's roughness by the registered 25% (best +12.6%), because a posterior mean conditioned on $\sim$800 points is smooth from the conditioning rather than the kernel—roughness and fit quality are not independently manipulable in a fitted GP.

Elimination 7: it is not distance to the data, and it is not surrogate inflation. The next hypothesis is that the ensemble's optima sit further off-support, where its mean is free to invent maxima, and that it over-predicts them more. We pre-registered both limbs and instrumented every returned optimum with its 10-NN distance to the offline data and its over-prediction against the oracle (168 cells, 30 seeds, reproducing the incumbent grid bit-exactly at seed 0). Both come back null: the ensemble's optima are $+0.116$ $[-0.004, 0.333]$ further out and $+0.043$ $[-0.652, 1.085]$ more inflated, both intervals covering zero, and the distance estimate is one task (Branin-2D, $+0.743$; six of seven at zero to three decimals). The loss is not null: at those same optima the ensemble is worse in *true oracle value* by -1.406 $[-2.803, -0.375]$ in units of $\mathrm{sd}(y_{\mathcal{D}})$, excluding zero, and binned on distance it is worse in four of five bins and tied in the fifth. The landscape law behind those bins is real—across 5,040 optima, distance predicts both over-prediction (ρ=+0.758) and true loss (ρ=−0.818)—but it does not discriminate: the classes sit at the same median distance (0.87 against 0.86) and differ in outcome anyway. Off-support maxima are inflated and worthless; that is a property of the problem, not of the ensemble.

The second limb is uninformative rather than negative, and the distinction is ours to make. It proposed raising the GP's prior mean until its far field stopped reverting to the data mean. The manipulation does not move the quantity it targets: a constant above every observation shifts the far-field posterior mean by under 1% of the distance to it (R=0.009 against a registered 0.25), because the fitted lengthscales are of order the domain diameter and no point in the feasible cube is in a reversion regime. The one arm that does remove reversion carries $211\times$ the incumbent's held-out RMSE, and a surrogate that has stopped predicting is not evidence about reversion. A premise failed rather than a prediction surviving—which establishes that here any account of the GP's advantage resting on prior reversion is unavailable, not merely untested.

The gap survives seven controls. Seven candidate explanations now have a control behind them and none survives: σ, per-member width, query budget, held-out accuracy, mean roughness, premise coverage, and distance-to-support with surrogate inflation. The seventh adds a positive localization: the loss is real and sits at the returned optimum's oracle quality, while every landscape property we can construct to explain it is matched between the classes. What differs is *which* off-distribution maximizers a surrogate's mean admits: not how rough those means are, not how far out the optima lie, and not how much they over-predict. That is a diagnosis, not a mechanism with a positive causal test behind it, and we label it as such.

Gradient collapse on the audited engine. Our own released tuning sweep concluded that the ensemble's gradient collapse is an artifact a trust region repairs—a result absent from the prior analysis. We report it in full, including the corner where it fires against us. By the sweep's own 5% rule, the tasks whose collapse survives tuning number $2/7$ with neither correction applied, $0/7$ under target scaling alone, $5/7$ under the protocol correction alone, and $4/7$ on the audited engine. The collapse is genuine surrogate geometry on most tasks, driven by the candidate/oracle correction rather than target scaling—and on one corner not genuine at all.

Two corrections to the coverage analysis. First, the collapse is not gradient-specific. We registered that ensemble coverage would also be low on CMA proposals; it is. Under the matched protocol gradient and CMA proposals differ in out-of-distribution coverage by 0.010, against 0.117 under the unmatched one, because both pool every iterate and return the top 128 by surrogate LCB. The claim narrows to the ensemble paired with any aggressive selector, and we withdraw the gradient-specific framing by name. Second, the premise-coverage value of 0.97 in the prior analysis belongs to a scikit-learn exact GP, not the differentiable GP the grid scores, whose coverage averages 0.831 and falls to 0.38 on Styblinski-5D; both are reported with the model each belongs to. Coverage of $\{\mu - f \leq \beta\sigma\}$ is exactly the validity of the lower bound, which is what makes these frequencies worth measuring.

6 Design-Bench Results

Surrogate main effect and the omnibus. Across all four engine corners the Design-Bench surrogate main effect is essentially zero—η^2_{surr} of 0.001, 0.032, 0.002 and 0.018 on five tasks, every point estimate at the floor of its own bootstrap interval—and the Friedman omnibus fails to reject at $p = 0.19$ to 0.76, with η^2_{opt} at 0.050–0.173. We therefore report no detectable difference at this power, never equivalence, because we are below the threshold for the test we ran: this omnibus is recommended for more than ten datasets (Demšar 2006), at $n{=}7$ a paired test needs $|d_z| \geq 1.27$ for 80% power at $\alpha{=}0.05$—larger than anything we observe—and a failure to reject is not a demonstration that an effect is absent (Agarwal et al. 2021). Two robustness notes belong with the number. Dropping GFP leaves η^2_{surr} at 0.001–0.021 (η^2_{opt} 0.047–0.184) with the omnibus still not rejecting. Adding the two MuJoCo tasks raises it only to 0.044–0.091 but does make the omnibus reject in three of four corners ($p = 0.011$ to 0.014)—and the decomposition localizes what rejects. Side by side in those three corners, η^2_{surr} is 0.091, 0.044, 0.062 against an η^2_{opt} of 0.146, 0.197, 0.181: the surrogate effect stays at the floor while the optimizer effect runs 1.6 to 4.5 times larger. The cross-corner pattern settles it—the one corner that does *not* reject ($p{=}0.16$) carries the *larger* surrogate effect (0.053, against a rejecting corner's 0.044), so surrogate effect size does not track rejection, while η^2_{opt} tracks it cleanly (0.096 where the omnibus fails, 0.146–0.197 where it rejects). Our limit: within any one corner the two intervals overlap, so this localizes point estimates across corners rather than separating the axes inside one. We also disclose a pooling defect in the prior analysis: its normalization ranged over eleven cells, two of them arms the grid never defines. Because mean-rank conclusions depend on the pool (Benavoli, Corani, and Mangili 2016), we unify to the nine defined cells before any rank claim.

Robustness to oracle approximation. We pre-registered this competing mechanism and the test that would kill it. The null survives on the exact-oracle subset in every corner ($p = 0.34$ to 0.51), while the only rejections anywhere are on the random-forest-oracle subset ($p = 0.02$ to 0.05 in three of four corners). Repeated oracle evaluation gives a variance at machine precision ($\leq 2.3 \times 10^{-13}$, and 3.6×10^{-15} on four of five tasks), so the null is not noise drowning a signal. The counter-objection is ours to state: with two exact-oracle tasks this subset test has very little power, so the result is consistent with the null rather than a powered acceptance, corroborated by $\eta^2_{\text{surr}} \approx 0$ and the noise floor.

Frozen cells at the dataset reference. Part of this null is structural rather than statistical. On TF-Bind-8, four of nine cells return exactly 1.0 on all 16 seeds with zero variance—the three exact-GP cells and ensemble-with-perturbation. That value is not a score: scores are min–max normalized to the offline dataset's own range, whose maximum is attained by seven tied rows already in the data, so a cell reporting 1.0 returned nothing better than its input. On Ant, two of three GP cells return the bit-identical constant 1.5287419557571411 with standard deviation exactly 0.0 across all sixteen seeds, and a third (SVGP with perturbation) hits it on ten of sixteen. Three cells across two surrogate classes converging on one bit-identical value is retrieval, not coincidence.

The mechanism label is the opposite of the one we had. Decode snap-back—movement in a relaxed logit space that argmax-decodes back to the starting design—predicts that an exact-constant freeze requires a decode step to snap through. Ant is continuous with no argmax decode anywhere in its path, so our pre-registered kill against that account fires directly. The surviving mechanism is LCB paralysis: the GP's lower confidence bound is locally maximal at the data, so the optimizer never leaves—a freeze occurring even without a decode step. This is the offline instance of a known reading of UCB-style acquisitions as local search (Fan et al. 2024), applied rather than discovered.

The Design-Bench inversion and the synthetic null are one finding. This section's most awkward number is that the optimizer axis *inverts* here—perturbation leads in all four corners (0.76–0.81 against gradient's 0.40–0.68) and η^2_{opt} runs two to three times η^2_{surr}—while the synthetic grid of Section 4 calls the optimizer negligible. Read through the frozen cells, the two suites stop disagreeing. If the GP cannot be moved, every optimizer scores it identically, the between-optimizer variance that survives comes almost entirely from the unfrozen ensemble cells, and the axis inverts as an arithmetic consequence of the freeze rather than as a substantive contradiction between benchmarks. A frozen surrogate reports the optimizer's variance because its own is gone—the payoff of a decomposition over a leaderboard. Four limits are ours. It is two of three Ant GP cells, not three—the gradient cell returns 1.3242 ± 0.1975 and is not frozen. Fact and label are separable, so losing the label does not lose the frozen cells. Frozen cells and genuinely near-tied cells are reported separately, never pooled. And the optimizer half of this section stays provisional: budget matching ran on the synthetic tasks only, so Design-Bench carries Confound 5 unremoved on exactly the axis carrying the effect here, and we do not promote it on a synthetic-only arm.

7 Discussion and Limitations

Pre-registered predictions and their outcomes. Four registered predictions came back against us and two registered mechanism arms failed to deliver what they were built for; we report all six, because undisclosed they would read as selective reporting to anyone who opens the repository. Our registered headline was that the acquisition optimizer explains most of the reported gap: it does not, η^2_{opt} being 0.038 even after correcting the budget imbalance that suppressed it, and the pre-committed contingency—pivot to the mechanism rather than quietly restate—is what this paper does. The ensemble-size reversal criterion did not fire, but its accompanying prediction confirmed: the headline is reported at the K that maximizes it (Confound 3). The β–σ criterion fired outright, narrowing Confound 4 to its per-task form, and the gradient-specific coverage framing was retracted on our own registered test. The two mechanism arms are the most expensive: smoothing the ensemble's mean was built to confirm the account we had published and refuted it, and the phantom-maxima probe was built to establish a mechanism and returned a seventh elimination.

Interpreting the optimizer effect. Under a matched surrogate-query budget of $Q{=}51{,}456$, $\eta^2_{\text{opt}} = 0.038$ $[0.003, 0.123]$, so optimizer choice explains little of the variance in this grid. Three qualifications travel with it. It replaces a published 0.005 that understates by roughly eightfold, a correction we disclose. The confirmation is not comfortable: the upper bound of 0.123 lies inside our pre-registered inconclusive band, so an effect up to about 0.12 is not excluded at $n{=}7$—the same precision limit that makes the corner decomposition unresolvable. And the secondary budget level does not corroborate cleanly (0.066 $[0.014, 0.340]$ at $Q{=}4{,}352$), so the null is established at high budget and underpowered at low, never unqualified. Most importantly, the *best* optimizer changes with budget: gradient leads at low budget (0.731) and perturbation at high (0.732), where the unmatched grid showed gradient modestly best throughout—an $11.8\times$ imbalance hiding itself by making one optimizer look mildly best everywhere. A small η^2_{opt} therefore licenses "optimizer choice explains little variance" and never "optimizer choice is arbitrary"—the second reading is false on our own data.

Scope of the optimizer-axis claim. We make no field-reversal claim. That the choice of statistical model often matters more than the acquisition heuristic is the organizing doctrine of the standard Bayesian-optimization review (Shahriari et al. 2016) and is a decade old; it governs online BO and the choice of acquisition *function family*, not the numerical search routine in the offline setting. What our design falsifies is the local premise of one recent paper—that offline black-box optimization has focused on improving surrogate models while using fixed search strategies (Chemingui et al. 2024)—by holding one shared protocol and varying both axes. We make no claim about the field's general beliefs about surrogates versus acquisitions. Nor do we claim ensemble size as a discovered mechanism (Li, Rudner, and Wilson 2024; Abe et al. 2022), equivalence on Design-Bench (Demšar 2006; Agarwal et al. 2021), or distance-aware uncertainty as ours (Liu et al. 2020; Fan et al. 2024).

Limitations. Seven synthetic tasks is a small n, binding the corner decomposition, the optimizer interval and the Design-Bench omnibus alike. The synthetic datasets are drawn once at seed 0, so reported variance is training and optimization variance, not data-draw variance. GFP is quarantined from coverage claims because its decode is degenerate, and we report that both ways: excluding it, one claim flips—the exact GP's mean in-distribution coverage moves from below nominal to above it—while the effect-size null is unchanged. Budget matching covers the synthetic tasks only, which is why the optimizer half of Section 6 stays provisional. And the surviving mechanism statement is a diagnosis with seven eliminations behind it and no positive causal test in front of it. The one positive account we pre-registered—off-support phantom maxima under prior reversion—was refuted on its first limb and untestable on its second, because at this operating point the GP is nowhere in a reversion regime inside the feasible set.

8 Conclusion

The reported advantage of GP-LCB in offline MBO is measured through five confounds specific to this comparison, and removing the two protocol confounds moves the surrogate main effect from a published 0.37 to 0.405 $[0.290, 0.556]$—against the direction audits usually run, with no precedent we could find, and the net of two corrections against a third. That decomposition is not resolvable at seven tasks and we say so; the firmer result is that any $\eta^2_{\text{surrogate}}$ here is a joint artifact of four operating-point coordinates papers do not state. What the advantage *is* remains open, narrowed by seven controls: not σ, width, budget, held-out accuracy, mean roughness, premise coverage, distance-to-support, or surrogate inflation. The loss itself is measured—the ensemble's optima are worse in true oracle value at matched distance and matched inflation. What we offer is the taxonomy, the protocol, and the engine stamp that makes either reproducible.

Ethical Statement

This work studies optimization methodology on public benchmarks and introduces no new data or deployed system.

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